

YEAR 7 MATHEMATICS · CHAPTER 11

Probability

11.01 Sample space**11.03** The probability scale**11.05** Complementary events**11.02** Probability**11.04** Experimental probability

Name _____

Class _____

Teacher _____

How to use this booklet

- Each lesson starts with **learning intentions** and **success criteria**. Write the **key terms** as your teacher explains them, and use the **notes** box.
- Work through the levels in order: Level 1, then Level 2, then Level 3. Extension questions stretch you further.
- Some questions ask you to **draw** a diagram, graph or construction. Use the space given, a ruler and a sharp pencil.
- At the end of each lesson, **show off your skill**: choose one question, then tear off the ticket and hand it in.
- Missed a lesson? Each lesson ends with a **summary** that has a worked example for every level. Start there, then try the questions.

WE DO

Grey boxes with a dark border. Work with your teacher. Copy what they write on the board.

YOU DO

Blue boxes. Work on your own. Show your working in the box and write your answer.

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In this chapter you will

- list the sample space of an experiment
- find the probability of simple events as fractions, decimals and percentages
- use the probability scale and chance words
- compare experimental and theoretical probability
- find the probability of complementary events

NSW Mathematics K–10 Syllabus (2022), Stage 4 Statistics and probability: MA4-PRO-C-01 solves problems involving the probabilities of simple chance experiments.



Acknowledgement of artwork

The artwork used throughout this booklet series was created by **Marni Tuala**. We thank Marni for sharing this artwork with our school community.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
11.01

Sample space

Date _____

Learning intentions

We are learning to:

- list the sample space of a chance experiment
- decide whether outcomes are equally likely

Success criteria

I can:

- ☐ list every possible outcome
- ☐ write the sample space in braces { }
- ☐ explain why outcomes are or are not equally likely

Key terms Write the definition as your teacher explains it.**Outcome****Sample space****Equally likely**

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

List the sample space for tossing a coin.

Example 2

List the sample space for rolling a die.

YOU DO

On your own: Set A. List the sample space.

1 Spinning a spinner with A, B, C, D

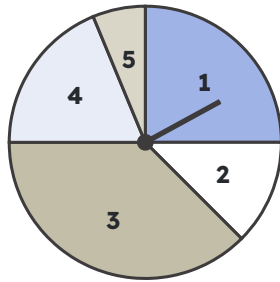
2 Choosing a day of the week

3 Choosing a vowel

4 Rolling a die

5 The result of a soccer match for one team

6 The season of a person's birthday

YOU DO On your own: Set B. Use the spinner.

- 1 How many outcomes?
- 2 Why aren't they equally likely?
- 3 Which number is most likely?
- 4 Which is least likely?
- 5 Which number has a probability of 25%?

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** List the outcomes of a driving test.
Example 4 List the classifications of a new film.
YOU DO On your own: Set C. List the sample space. Are the outcomes equally likely?

- 1 The public transport students use to get to school

- 3 Choosing a letter from the word APPLE

- 5 The colour of a traffic light when you arrive

- 2 Choosing a letter from the word MATHS

- 4 The number of heads when 2 coins are tossed

- 6 Drawing a card suit from a deck



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Set B spinner: how many outcomes?
Are they equally likely?
Example 6 List the sample space for the winner
when Katrina plays Biljana at tennis.
YOU DO On your own: Set D. Two-step experiments.**1** List the sample space for tossing 2 coins (e.g. HT).
3 How many outcomes are there for rolling 2 dice?
Explain.
2 List all outcomes for spinning spinner ABCD and
tossing a coin.
4 Design a spinner with 4 outcomes that are not
equally likely.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show
your working.

Name _____ Class _____

Level 1List the sample space for choosing
a primary colour.
Level 2How many outcomes are there
when rolling a die?
Level 3Are the outcomes of a test
(pass/fail) equally likely? Explain.

Extension · Stretch yourself

Worked example

How many outcomes are there when 3 coins are tossed? List them.

- Each coin has 2 outcomes: $2 \times 2 \times 2$
- HHH, HHT, HTH, HTT, THH, THT, TTH, TTT

Answer: 8 outcomes

YOU DO

On your own: read the worked example, then try these.

- E1** How many outcomes when 4 coins are tossed?

- E2** How many different 2-letter codes can be made from A, B, C if letters can repeat?

Lesson summary

Missed the lesson? Start here.

Remember

- An **outcome** is one possible result.
- The **sample space** is the list of all outcomes.
- Write it in braces: {H, T}.
- Outcomes are **equally likely** if each has the same chance.

Level 1

Tossing a coin

- 2 outcomes

Answer: {H, T}

Level 2

Rolling a die

- 6 outcomes

Answer: {1, 2, 3, 4, 5, 6}

Level 3

Spinner with unequal sectors

- Different sector sizes

Answer: not equally likely

Extension

3 coins

- $2 \times 2 \times 2$

Answer: 8 outcomes

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LESSON
11.02

Probability

Date _____

Learning intentions

We are learning to:

- find the probability of a simple event

Success criteria

I can:

- ☐ use $P(\text{event}) = \frac{\text{number of favourable outcomes}}{\text{total outcomes}}$
- ☐ write probability as a fraction, decimal or percentage
- ☐ round percentages to 1 decimal place

Key terms Write the definition as your teacher explains it.

Probability

Event

At random

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** On a die, find $P(1)$.**Example 2** On a die, find $P(\text{odd number})$.**YOU DO** On your own: Set A. A die is rolled. Find the probability of rolling:

1 a 6

4 a 7

2 an even number

5 a number greater than 0

3 a number less than 5

6 a 2 or a 3

YOU DO

On your own: Set B. A letter is chosen at random from PROBABILITY. Find the probability of choosing:

1 B

4 Y

2 I

5 a letter in the word ROB

3 a vowel

6 Z

Level 2 · Building up**WE DO**

With your teacher: copy what they write on the board.

Example 3On a die, find $P(\text{number less than 5})$.**Example 4**A box has 15 dark, 13 milk and 12 white chocolates. Find $P(\text{white})$.**YOU DO**

On your own: Set C. Find each probability as a fraction and a percentage (1 d.p.).

1 15 dark, 13 milk, 12 white chocolates: $P(\text{white})$ 3 $P(\text{not milk})$ 5 $P(\text{exercise books or paper})$ 2 $P(\text{dark})$ 4 Truck (1040 boxes): $P(\text{rulers})$ 6 $P(\text{erasers})$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** A truck has 325 boxes of books, 210 of rulers, 360 of paper and 145 of pens. Find $P(\text{rulers})$ as a % (1 d.p.).
Example 6 Using the truck, find $P(\text{no pens})$ as a % (1 d.p.).
YOU DO On your own: Set D. Problems.

- 1 A class has 14 girls and 11 boys. One student is chosen at random. Find $P(\text{boy})$ as a decimal.

- 3 A bag has red and blue counters. $P(\text{red}) = 0.35$. What is $P(\text{blue})$?

- 2 A raffle sells 250 tickets. You buy 4. Find your probability of winning as a percentage.

- 4 A bag has 12 counters and $P(\text{green}) = \frac{1}{4}$. How many are green?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1Find $P(5)$ on a die.
Level 2A bag has 3 red and 7 blue marbles. Find $P(\text{red})$.
Level 3

Write your answer to Question 2 as a percentage.

Extension · Stretch yourself

Worked example

A card is drawn from a standard deck of 52.
Find $P(\text{red king})$.

- There are 2 red kings (hearts, diamonds)
- $\frac{2}{52}$ simplifies

Answer: $\frac{1}{26}$

YOU DO

On your own: read the worked example, then try these.

- E1** Find $P(\text{a heart or a king})$ from a standard deck.

- E2** Find $P(\text{a picture card: J, Q, K})$ from a standard deck.

Lesson summary

Missed the lesson? Start here.

- Remember**
- $P(\text{event}) = \text{favourable outcomes} \div \text{total outcomes}$.
 - Simplify fractions.

- Probabilities are between 0 and 1.
- Percentage: $\times 100\%$, round as asked.

Level 1

$P(\text{odd})$ on a die

- 3 odd of 6

Answer: $\frac{1}{2}$

Level 2

$P(\text{white chocolate})$

- 12 of 40

Answer: $\frac{3}{10}$ (C)

Level 3

$P(\text{rulers})$

- $210 \div 1040$

Answer: 20.2%

Extension

$P(\text{red king})$

- 2 of 52

Answer: $\frac{1}{26}$

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LESSON
11.03

The probability scale

Date _____

Learning intentions

We are learning to:

- describe chance using words and the probability scale

Success criteria

I can:

- ☐ place events on a scale from 0 (impossible) to 1 (certain)
- ☐ match probabilities to words such as unlikely and good chance
- ☐ find probabilities from a jar of objects

Key terms Write the definition as your teacher explains it.

Certain

Impossible

Even chance

Notes

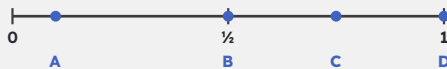
Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

Where on the scale is the chance that the next baby is a girl?



Example 2

Which word describes a probability of 0.29?

YOU DO

On your own: Set A. Describe each event using a chance word.

- 1 The sun will rise tomorrow.

- 4 It will snow in Sydney in January.

- 2 Rolling a 7 on a die.

- 5 Rolling a number less than 6.

- 3 A coin lands on heads.

- 6 Your teacher is over 10 years old.

YOU DO

On your own: Set B. Match each probability to a word: impossible, unlikely, even chance, likely, certain.

1 0

4 0.85

2 0.29

5 1

3 $\frac{1}{2}$

6 15%

**Level 2 · Building up****WE DO**

With your teacher: copy what they write on the board.

Example 3

Which is certain: choosing a 4 from {1, 2, 3, 4}; a red ball from green balls; an odd or even number on a die; a baby being a girl?

Example 4

A jar has 1 red, 6 yellow, 2 white and 5 black jellybeans. Find P(yellow).

YOU DO

On your own: Set C. A jar has 1 red, 6 yellow, 2 white and 5 black jellybeans. Find:

1 P(yellow)

3 P(blue)

5 P(black)

2 P(white or red)

4 P(not green)

6 P(not yellow)

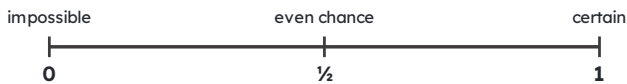


Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Find P(white or red).
Example 6 Find P(blue) and P(not green).
YOU DO On your own: Set D. Mark each event on the scale.

- 1 Mark: A rolling an even number; B rolling a 6; C rolling a number from 1 to 6; D rolling a 0.



- 2 Describe an event that is: (a) impossible (b) unlikely (c) certain.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

What is the probability of an impossible event?

Level 2Which word describes $P = 0.9$?
Level 3

A bag has 3 red and 5 blue. Find P(not red).

Extension · Stretch yourself

Worked example

A bag has 20 counters. $P(\text{red}) = 0.4$ and $P(\text{blue}) = 0.35$. The rest are green. How many are green?

1. $P(\text{green}) = 1 - 0.4 - 0.35 = 0.25$
2. 0.25×20

Answer: 5

YOU DO

On your own: read the worked example, then try these.

- E1** A spinner has $P(\text{red}) = \frac{1}{3}$. It has 12 equal sectors. How many are red?

- E2** Design a bag of 10 marbles where $P(\text{blue})$ is “unlikely” and $P(\text{red})$ is “likely”.

Lesson summary

Missed the lesson? Start here.

Remember

1. Probability goes from 0 (impossible) to 1 (certain).
2. $\frac{1}{2}$ is an even chance.
3. Less than $\frac{1}{2}$: unlikely. More than $\frac{1}{2}$: likely.
4. Count the objects to find each probability.

Level 1

Baby is a girl

1. About half

Answer: B (even chance)

Level 2

0.29

1. Less than $\frac{1}{2}$

Answer: unlikely

Level 3

$P(\text{yellow})$

1. 6 of 14

Answer: $\frac{3}{7}$

Extension

Green counters

1. 0.25×20

Answer: 5

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LESSON
11.04

Experimental probability

Date _____

Learning intentions

We are learning to:

- compare experimental and theoretical probability

Success criteria

I can:

- ☐ find theoretical probability from equally likely outcomes
- ☐ find experimental (observed) probability from results
- ☐ find expected frequency = probability $\times$ number of trials

Key terms Write the definition as your teacher explains it.

Experimental probability

Expected frequency

Trial

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** A coin is tossed 80 times. What is the theoretical $P(\text{head})$?**Example 2** What is the expected number of heads in 80 tosses?**YOU DO** On your own: Set A. Find the theoretical probability.1 $P(\text{head})$ on a coin
2 $P(4)$ on a die
3 $P(A)$ on spinner ABCD
4 $P(\text{even})$ on a die
5 $P(\text{red})$ from 3 red, 2 blue
6 $P(\text{vowel})$ from A, B, C, D, E

YOU DO On your own: Set B. Find the expected frequency.

- | | | | |
|----------------------------|----------------------|--|----------------------|
| 1 Heads in 80 tosses | <input type="text"/> | 4 A on spinner ABCD in 200 spins | <input type="text"/> |
| 2 6s in 60 rolls | <input type="text"/> | 5 Red in 100 draws (3 red, 2 blue, replaced) | <input type="text"/> |
| 3 Even numbers in 50 rolls | <input type="text"/> | 6 Tails in 1000 tosses | <input type="text"/> |

Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

Example 3 It landed heads 24 times. What is the observed probability of a head?

Example 4 How does the observed frequency compare with the expected frequency?

YOU DO On your own: Set C. Find the observed (experimental) probability.

- | | | |
|-------------------------|------------------------|------------------------|
| 1 24 heads in 80 tosses | 3 42 reds in 100 draws | 5 130 A's in 400 spins |
| <input type="text"/> | <input type="text"/> | <input type="text"/> |
| 2 15 sixes in 60 rolls | 4 7 wins in 20 games | 6 0 fives in 12 rolls |
| <input type="text"/> | <input type="text"/> | <input type="text"/> |



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** A die is rolled 60 times. How many 6s are expected?
Example 6 Why do more trials give a better estimate?
YOU DO On your own: Set D. Do an experiment. Toss a coin 20 times (or use results from your class).

1 Record your results in a tally table.

Outcome	Tally	Frequency
Heads		
Tails		

2 Find your observed $P(\text{head})$. Compare it with the theoretical probability. Combine with the class results. What do you notice?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1What is the theoretical $P(3)$ on a die?
Level 2

How many 3s are expected in 120 rolls?

Level 3A 3 came up 25 times in 120 rolls. Find the observed $P(3)$.

Extension · Stretch yourself

Worked example

A spinner landed on red 45 times in 150 spins. Estimate how many times it will land on red in 1000 spins.

1. Observed $P(\text{red}) = 45 \div 150 = 0.3$
2. 0.3×1000

Answer: about 300 times

YOU DO

On your own: read the worked example, then try these.

- E1** A drawing pin landed point up 36 times in 60 throws. Estimate how many times it lands point up in 250 throws.

- E2** Is this spinner likely to have equal sectors if there are 4 colours? Explain.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Theoretical $P = \text{favourable} \div \text{total}$ (equally likely outcomes).
 3. Expected frequency = $P \times \text{number of trials}$.

2. Observed $P = \text{number of times it happened} \div \text{number of trials}$.
4. More trials: observed P gets closer to theoretical P .

Level 1

Theoretical $P(\text{head})$

1. 1 of 2

Answer: $\frac{1}{2}$

Level 2

Expected heads in 80

1. $\frac{1}{2} \times 80$

Answer: 40

Level 3

Observed $P(\text{head})$

1. $24 \div 80$

Answer: $\frac{3}{10}$

Extension

Red in 1000 spins

1. 0.3×1000

Answer: about 300

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LESSON
11.05

Complementary events

Date _____

Learning intentions

We are learning to:

- find the probability of complementary events

Success criteria

I can:

- ☐ describe the **complement** of an event ("not" the event)
- ☐ use $P(\text{not } E) = 1 - P(E)$
- ☐ solve problems using complementary events

Key terms Write the definition as your teacher explains it.

Complementary events

Complement

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Write the complement of selecting a queen from a deck, and its probability.
Example 2 Write the complement of rolling a multiple of 3 on a die, and its probability.
YOU DO On your own: Set A. Write the complementary event.

1 Rolling a 6

2 Getting a head

3 Drawing a red card

4 Winning a game (no draws)

5 Choosing a vowel

6 Rolling an even number

YOU DO On your own: Set B. Find $P(\text{not } E)$.

1 $P(E) = \frac{1}{4}$

4 $P(E) = \frac{5}{12}$

2 $P(E) = 0.3$

5 $P(E) = 0.07$

3 $P(E) = 45\%$

6 $P(E) = 1$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Write the complement of buying the winning ticket out of 550 sold, and its probability.**Example 4** Names Ashleigh, Hamish, Sarah, Jake, Hayley: find $P(\text{Hamish})$.**YOU DO** On your own: Set C. Find each probability.

1 $P(\text{not a queen})$ from a deck

3 $P(\text{not winning})$ with 1 ticket of 550

5 $P(\text{not brown})$: 8 brown, 4 black, 3 white

2 $P(\text{not a multiple of 3})$ on a die

4 $P(\text{not Hamish})$ from the 5 names

6 $P(\text{not a heart})$ from a deck



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Find $P(\text{a girl's name})$ and $P(\text{not beginning with H})$.
Example 6 A bag has 8 brown, 4 black and 3 white marbles. Find $P(\text{not brown})$.
YOU DO On your own: Set D. Problems.**1** The probability it will rain tomorrow is 0.35. What is the probability it will not rain?
3 In a class, $P(\text{student has a pet}) = 17/25$. There are 25 students. How many do not have a pet?
2 $P(\text{a team wins}) = 0.6$ and $P(\text{draw}) = 0.1$. Find $P(\text{the team loses})$.
4 Explain why $P(E) + P(\text{not } E) = 1$.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write the complement of rolling a 2.

Level 2If $P(E) = 0.8$, find $P(\text{not } E)$.
Level 3A bag has 5 red, 3 blue and 2 green. Find $P(\text{not red})$.

Extension · Stretch yourself

Worked example

A spinner has $P(\text{red}) = 0.2$, $P(\text{blue}) = 0.45$, $P(\text{green}) = \text{the rest}$. Find $P(\text{not blue})$ and $P(\text{green})$.

- $P(\text{not blue}) = 1 - 0.45$
- $P(\text{green}) = 1 - 0.2 - 0.45$

Answer: $P(\text{not blue}) = 0.55$, $P(\text{green}) = 0.35$

YOU DO

On your own: read the worked example, then try these.

- E1** A bag has red, white and blue marbles. $P(\text{red}) = \frac{1}{3}$ and $P(\text{white}) = \frac{1}{4}$. Find $P(\text{blue})$.

- E2** In 200 spins of the spinner above, how many times would you expect NOT green?

Lesson summary

Missed the lesson? Start here.

- Remember**
- The complement of E is “not E”.
 - $P(\text{not E}) = 1 - P(E)$.

- E and not E cover all outcomes.
- Use it when “not” is easier to count.

Level 1

Not a queen

- 4 queens of 52

Answer: $\frac{12}{13}$

Level 2

Not a multiple of 3

- $1 - \frac{1}{3}$

Answer: $\frac{2}{3}$

Level 3

Not brown

- 7 of 15

Answer: $\frac{7}{15}$

Extension

Spinner

- $1 - 0.45$

Answer: 0.55

This space is left blank on purpose. It is the back of your exit ticket.